Who Loses and Why?

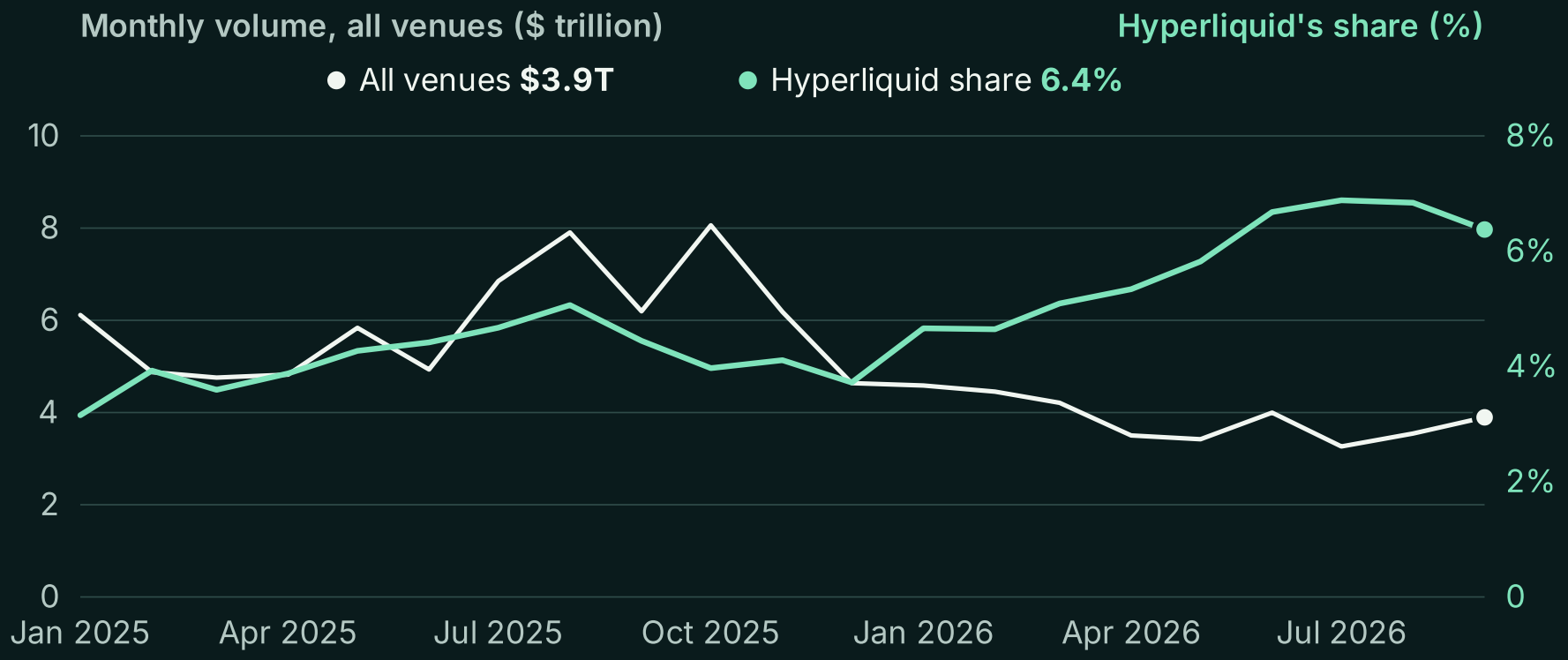
Market Microstructure of Perpetual Futures

Xinmeng Zeng · Ruizhe Jia *Stanford University*

Shouqiao Wang · Ciamac Moallemi *Columbia University*

Perpetual Futures Trade Trillions a Month

- **A perpetual future** is a futures contract with no expiry: traders go long or short, with leverage.
- **Coming onshore:** the first U.S. bitcoin perpetual was approved in 2026.



Hyperliquid Shows the Trader, Not Just the Trade

Top 10 coins · 2025.03–2026.06 · $2.95 trillion volume · 729,540 wallets

- **Every fill names the wallet**, its order type, and whether it opens, closes or is a liquidation.

TIME (UTC)	WALLET	COIN	SIDE	PRICE	SIZE	ORDER TYPE	ROLE	DIRECTION	LIQUIDATION
14:03:07.212	0x7c2…e41	SOL	Buy	180.03	$35,000	FrontendMarket	Taker	Open long	No
14:03:07.212	0x91a…0bd	SOL	Sell	180.03	$35,000	Alo	Maker	Open short	No
14:03:07.950	0x4de…772	ETH	Sell	4,102.5	$12,400	Ioc	Taker	Close long	No
14:03:08.105	0x2b8…c19	BTC	Sell	108,214	$61,800	Liquidation	Taker	Close long	Yes

Three Groups Trade Very Differently



Classification: Retail trades via frontend

HYPERLIQUID

Replay · real 5-minute BTC candles and order-book snapshots · book from Hyperliquid data

↻ REPLAY

BTC-USDC 20x

Mark	24h Change	24h Volume
110,592	-1.90%	$3,142,527,612

5m · Oct 11, 23:55 UTC



Order Book BTC

Price	Size (BTC)	Total (BTC)
110,600	0.1355	1.4839
110,599	0.9033	1.3484
110,597	0.0002	0.4451
110,595	0.0972	0.4449
110,593	0.3477	0.3477
Spread	1	0.001%
110,592	7.6081	7.6081
110,591	0.9033	8.5114
110,590	0.0648	8.5761
110,589	0.0272	8.6034
110,588	1.8426	10.4460

Cross | 20x

Market | Limit | Pro

Buy / Long | Sell / Short

Available to Trade 1,000.00 USDC

Size 0.05 BTC

Size only: it fills at the asks, now. Sent from this page, the order is stamped FrontendMarket.

Buy / Long

Order Value $5,530

Fees 0.0450% / 0.0150%

Classification: Market makers post liquidity

ORDER BOOK · HYPE

PRICE	SIZE ($)	TOTAL ($)
100.03	$6,000	$30,000
100.02	$8,000	$24,000
100.01	$16,000	$16,000
spread		
99.99	$13,000	$13,000
99.98	$10,000	$23,000
99.97	$9,000	$32,000

Alo buy at 99.99

Accepted: it waits on the book as a maker.

Alo never crosses the spread.

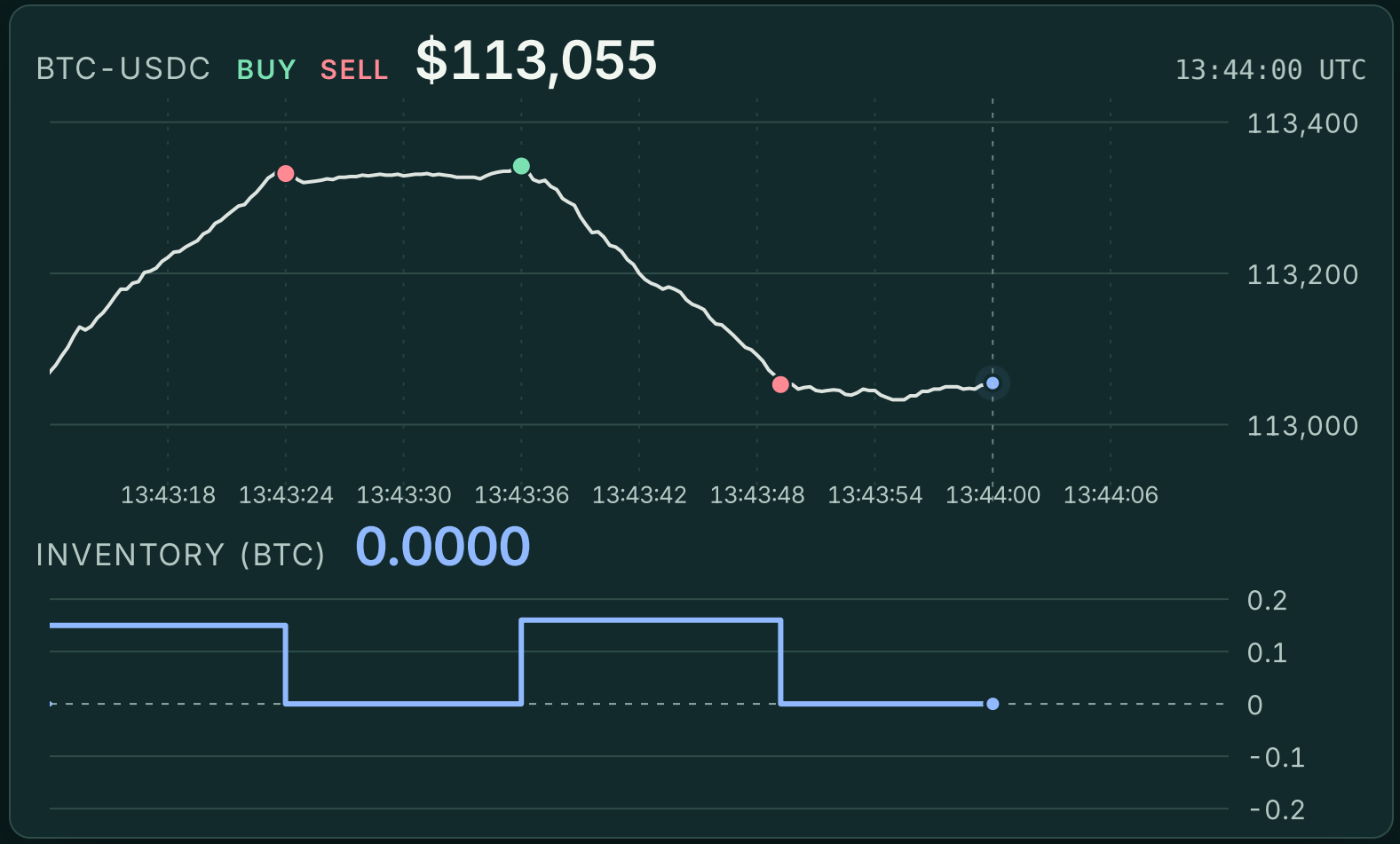
Classification: Algo traders are fast bots

strategy.py ● LIVE

```
signal = model.predict(book)
if signal.strong:
    api.ioc(side, qty)
for fill in api.fills():
    book.update(fill)
```

13:43:10.8 FILL BUY 0.15 @113,027
13:43:24.0 FILL SELL 0.15 @113,332
13:43:36.0 FILL BUY 0.16 @113,342
13:43:49.2 FILL SELL 0.16 @113,053

BTC 0.0000 P&L −$0

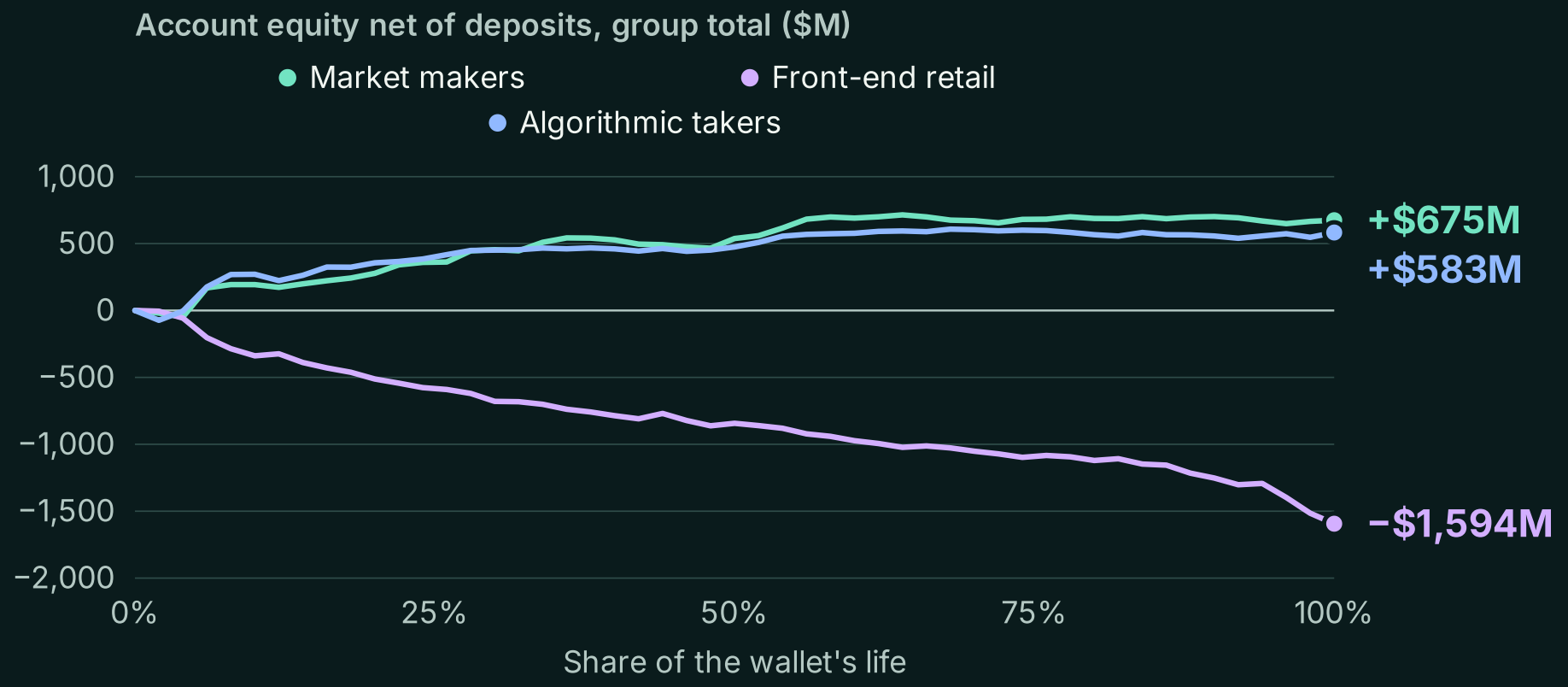


Three Groups Trade Very Differently

- **Retail start small, use high leverage** and trade mostly as takers with front-end market orders.

GROUP	TAKER RATIO	OPENING LEVERAGE, MEDIAN	STARTING EQUITY · P50 / P95
Market makers	20%	1.4×	$26,052 / $1.64M
Algorithmic takers	69%	3.7×	$386 / $53,485
Front-end retail	**85%**	**9.0×**	**$306 / $50,342**

Retail Accounts Lose $1.6 Billion Net of Deposits



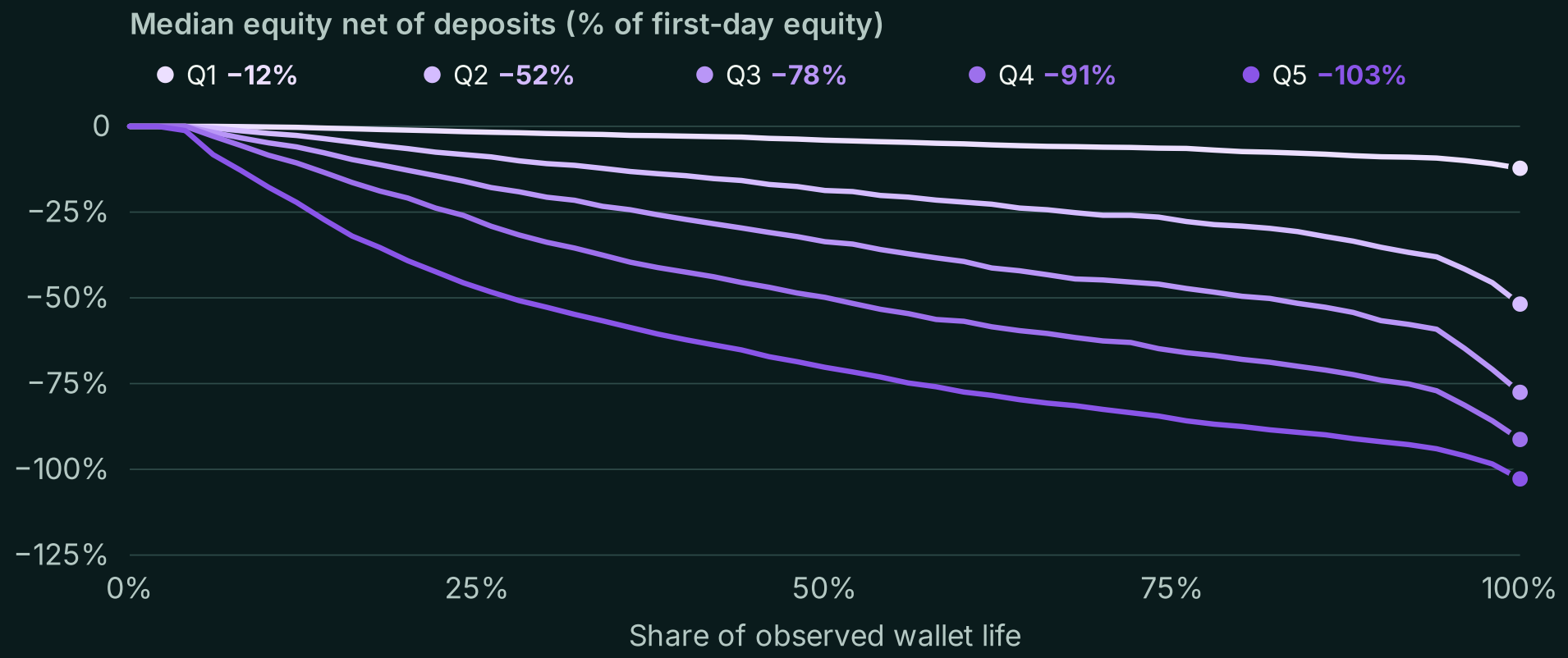
- **The typical retail wallet** loses two-thirds of what it starts with.

Which Retail Loses

- **High-leverage retail start with less equity** and lose it all.

RETAIL COHORT, MEDIAN WALLET	OPENING LEVERAGE	STARTING EQUITY	AVERAGE TRADE SIZE	CASH FLOW AT END OF LIFE
Q1 (lowest)	2.6×	$186	$250	−12%
Q2	5.5×	$195	$899	−52%
Q3	10.0×	$177	$1,105	−78%
Q4	15.8×	$99	$1,175	−91%
Q5 (highest)	**25.1×**	**$33**	$1,095	**−103%**

Gambler's Ruin: They Lose Their Stake and Deposit Again



- **The typical Q5 wallet** loses all it starts with, deposits again, and loses more.
- **Half of Q5 wallets** lose at least their stake, against one in six in Q1.

We Split Retail Loss into Three Channels

1. **Fees**
   what retail pay the venue

2. **Liquidation**
   positions the platform closes when equity runs out

3. **Trading loss**
   execution cost and timing of voluntary trades
   Measured by markout: how the price moves after the trade.

Markout: How the Price Moves After a Trade

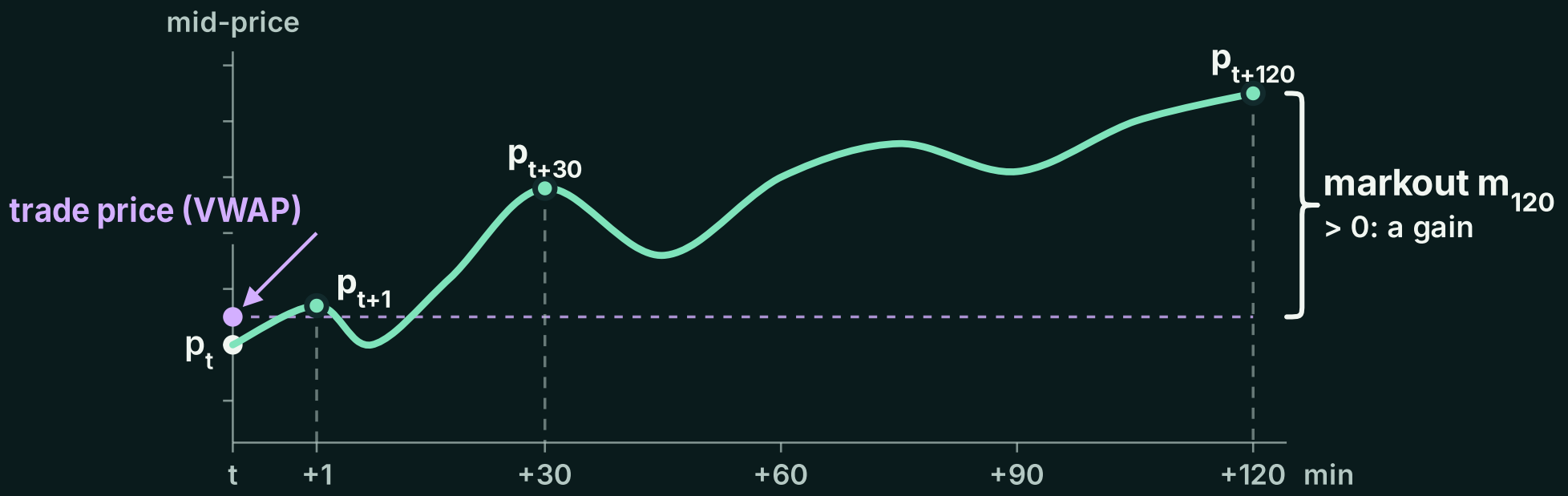
$$m_h = s \times \frac{p_{t+h} - \bar{p}_t}{\bar{p}_t} \times 10^4$$

m_h = markout, bps

s = +1 buy, −1 sell

$\bar{p}_t$ = trade price (VWAP) at time *t*

p_{t+h} = mid-price *h* minutes later



Retail Lose $1.11 Billion; the Venue Collects $830 Million in Fees

- **Market makers gain,** algorithmic takers break even, retail lose.
- **Fees are the largest channel** of the retail loss.

GROUP	TRADING LOSS	LIQUIDATION	FUNDING	FEES	TOTAL
Market makers	+$296M	−$38M	+$4M	−$50M	+$211M
Algorithmic takers	+$40M	+$11M	+$1M	−$55M	−$4M
Front-end retail	−$154M	−$226M	−$5M	**−$725M**	**−$1.11B**

Channel 1: Fees

1. **Fees**
   what retail pay the venue
2. Liquidation
   positions the platform closes when equity runs out
3. Trading loss
   execution cost and timing of voluntary trades

How the Fee Is Set: Staking Tier, Then Volume, Then Role

1 HYPE staked

The staking tier sets the discount on every rate.

2 Trading volume

Within the tier, more volume means a lower fee rate.

3 Taker or maker

Takers pay the higher rate. The largest makers earn a rebate.

VOLUME TIER	14-DAY WEIGHTED VOLUME	TAKER BASE RATE	MAKER BASE RATE
Tier 0	$0–$5M	4.5 bps (0.045%)	1.5 bps (0.015%)

Fees: Retail Pay 87% of Fees on 42% of Volume

- **Retail trade mostly as takers,** and takers pay the higher rate.
- **Retail sit in the highest fee tier:** little HYPE staked, low volume.

	TAKER FEE (BPS)	MAKER FEE (BPS)	TAKER RATIO	VOLUME FROM NON-STAKERS
Market makers	1.71	**rebate 0.22**	20%	24%
Algorithmic takers	2.87	0.55	69%	43%
Front-end retail	**3.32**	0.80	**85%**	**74%**

Channel 2: Liquidation

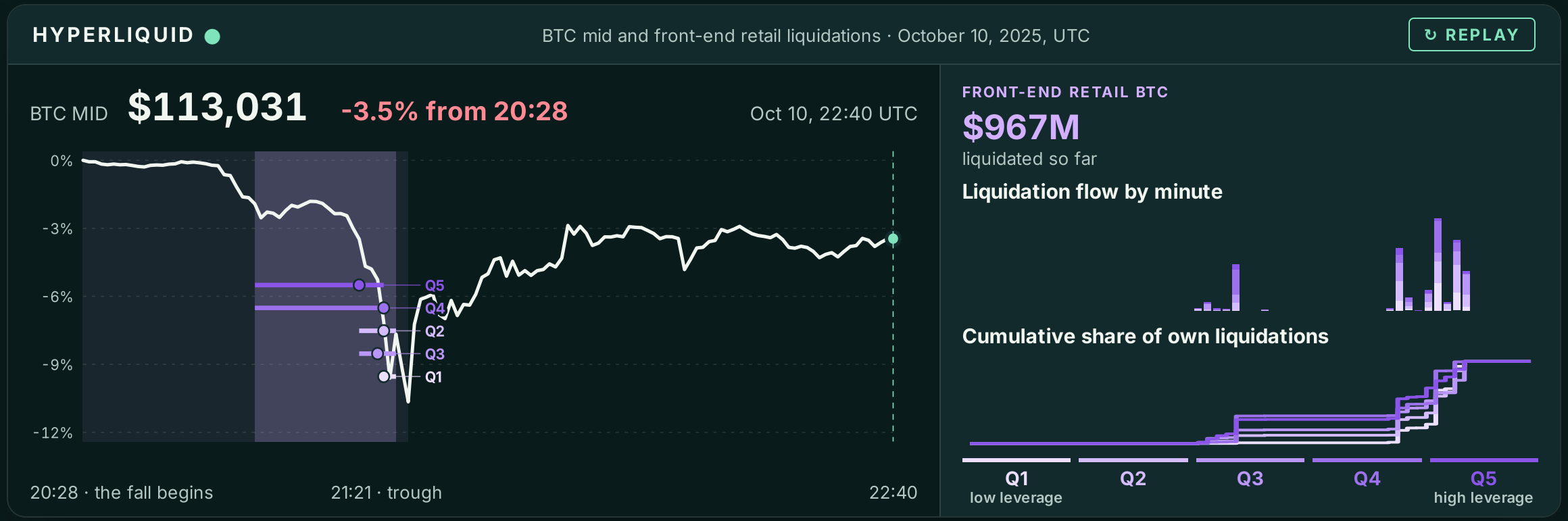
1. Fees
   what retail pay the venue
2. **Liquidation**
   positions the platform closes when equity runs out
3. Trading loss
   execution cost and timing of voluntary trades

Liquidation: Retail Bear It, and It Is the Costliest Exit

- **Retail bear almost all liquidations:** 97.6% of liquidation volume outside October 10.
- **Liquidations are large, pay the widest half-spread and have the worst timing:** they land near the bottom of a fall, and the price reverts soon after.

THREE WAYS TO CLOSE A TRADE	TRADE SIZE	HALF-SPREAD (BPS)	PRICE IMPACT (BPS)
Gain-realizing close	$7.0k	0.69	+2.54
Loss-realizing close	$7.7k	0.92	+0.66
Liquidation	**$31.1k**	**15.16**	**−28.48**

Low Leverage Is Liquidated Last, at the Worst Price



- **Q5 is liquidated often,** after small moves.
- **Q1 is liquidated rarely,** but at the worst price during the biggest cascades.

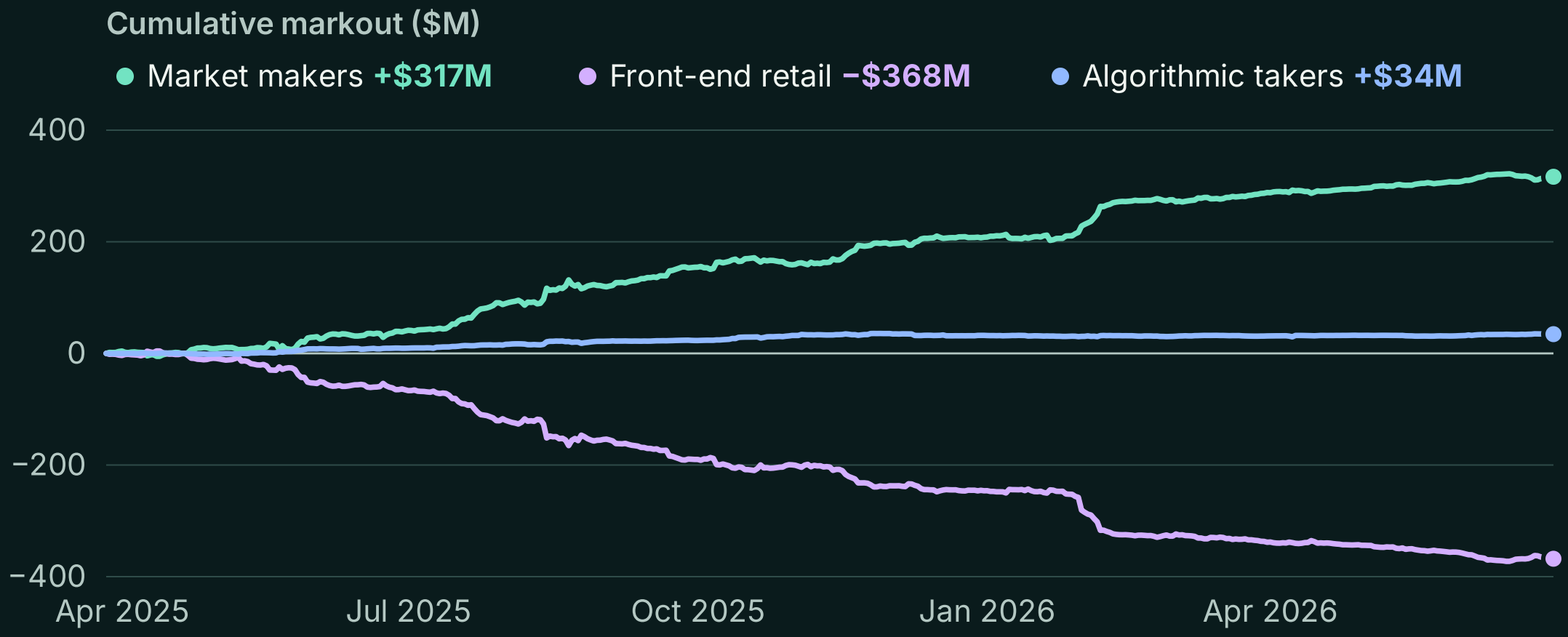
Channel 3: Trading Loss

1. **Fees**
   what retail pay the venue

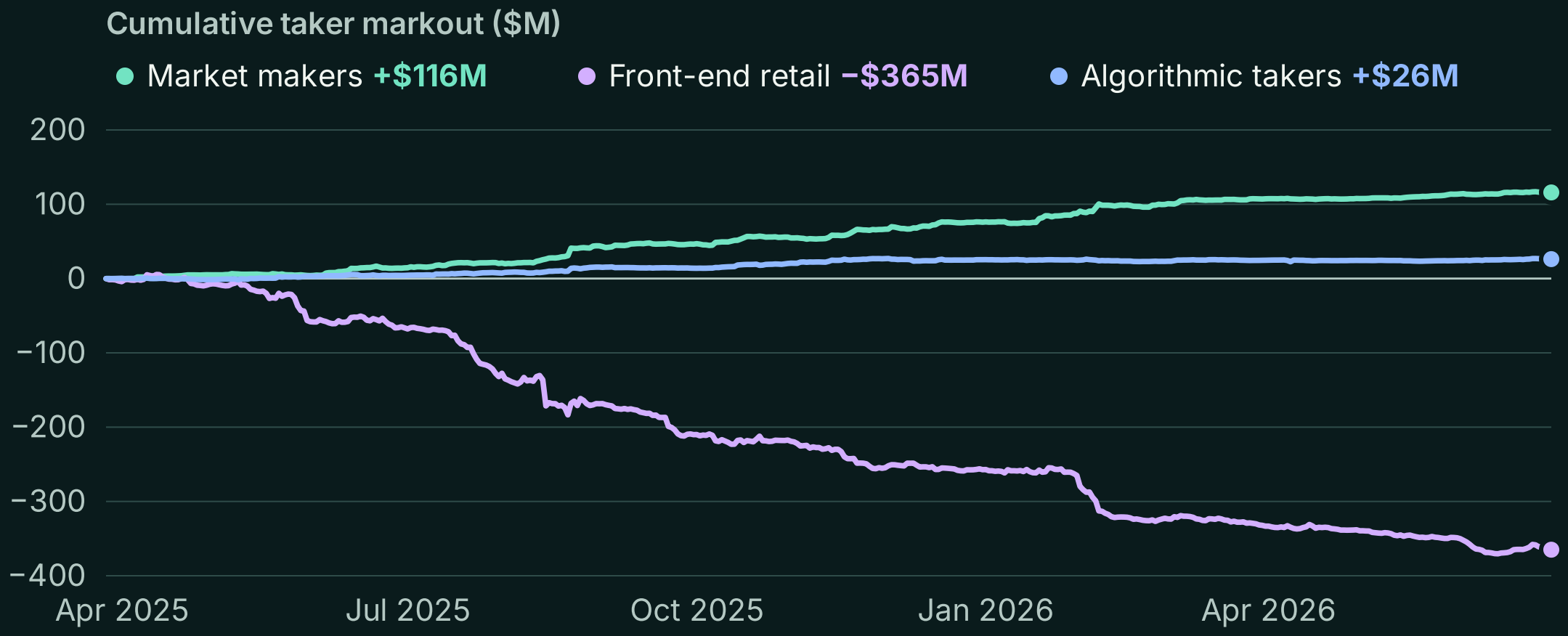
2. **Liquidation**
   positions the platform closes when equity runs out

3. **Trading loss**
   execution cost and timing of voluntary trades

Retail's Loss Accumulates Over Time

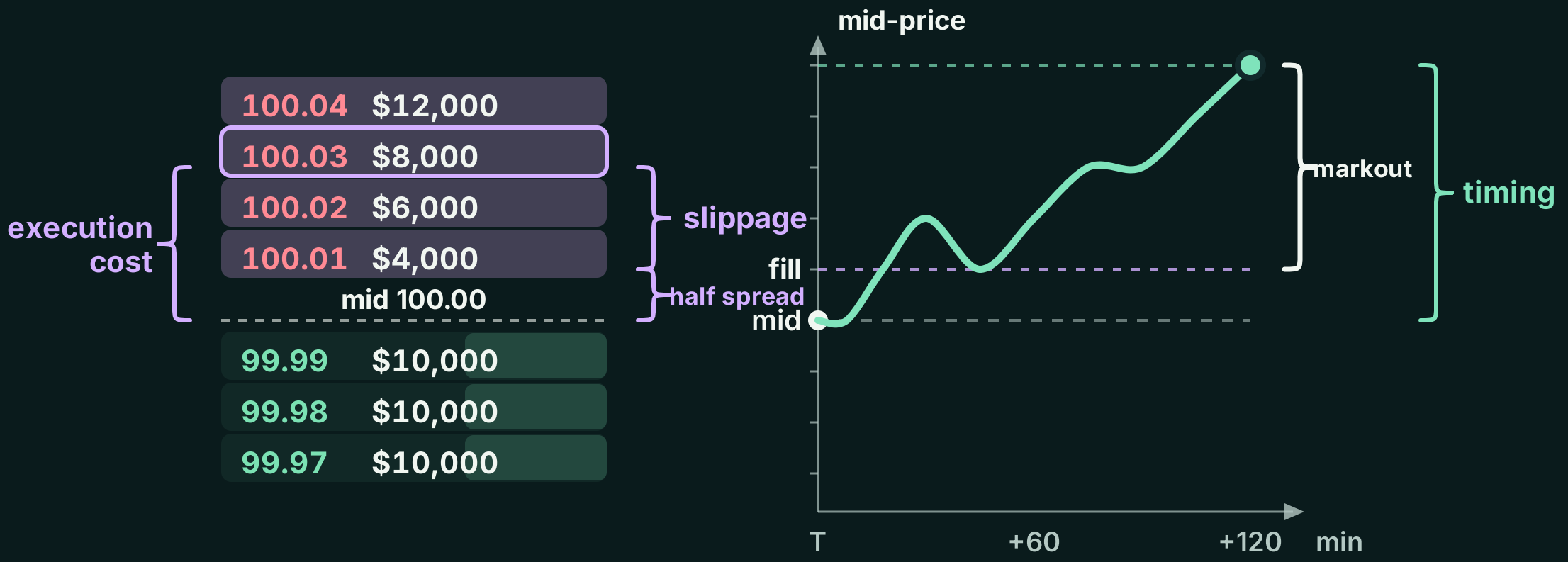


The Loss Is Concentrated on the Taker Side

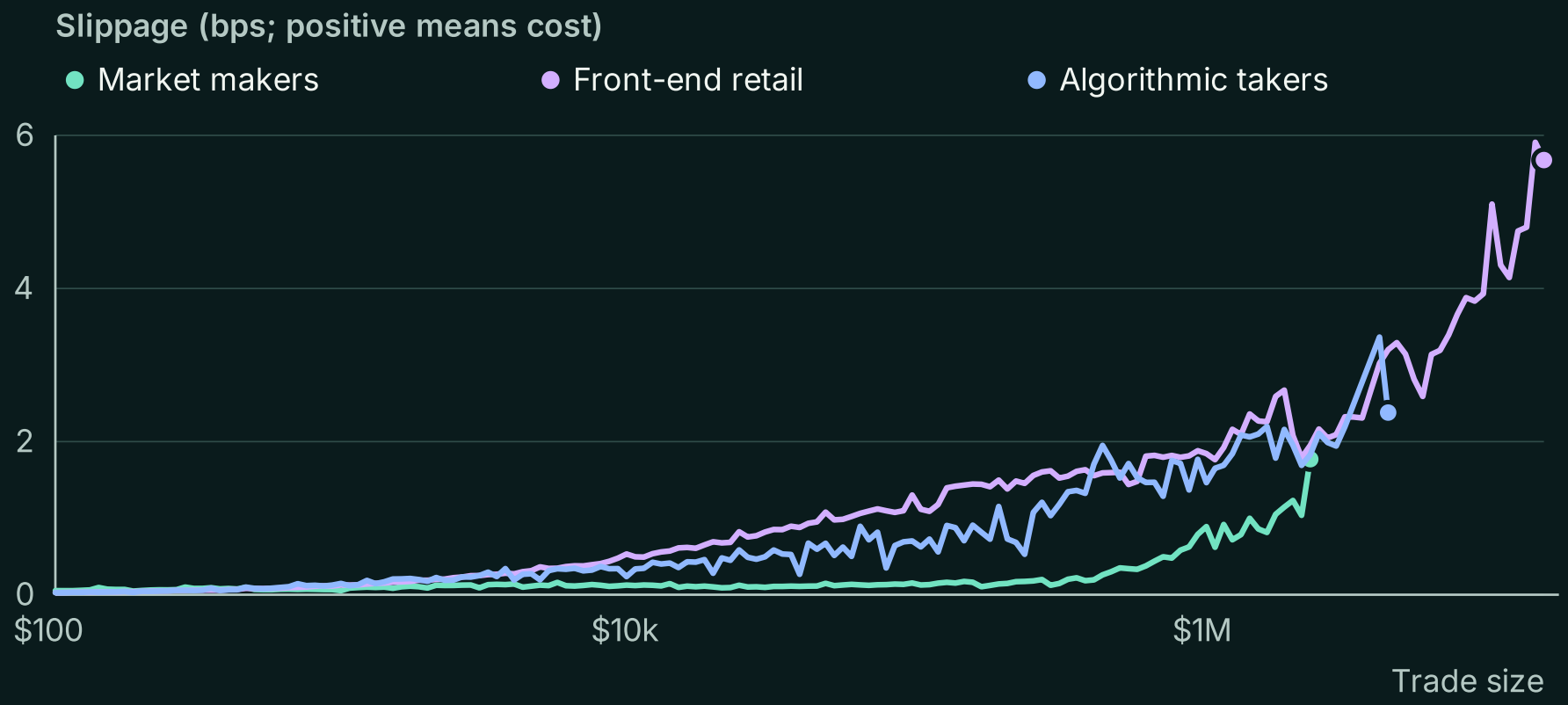


Taker Markout = Execution Cost + Timing

Taker markout = Execution cost + Timing | Execution cost = Half spread + Slippage



Execution Cost: Retail Walk the Book



- **Same half-spread for all;** retail pay in slippage.
- **Market makers control slippage** with loc limit orders; retail do not.

The Same Order Walks Further Through a Thin Book

THE SAME BUY OF $30,000 , AT TWO DIFFERENT MOMENTS

Front-end retail

a thin book

FrontendMarket order: size only

ASKS	SIZE
100.10	$16,000
100.05	$8,000
100.03	$6,000

Walks 3 levels: more slippage

Market maker

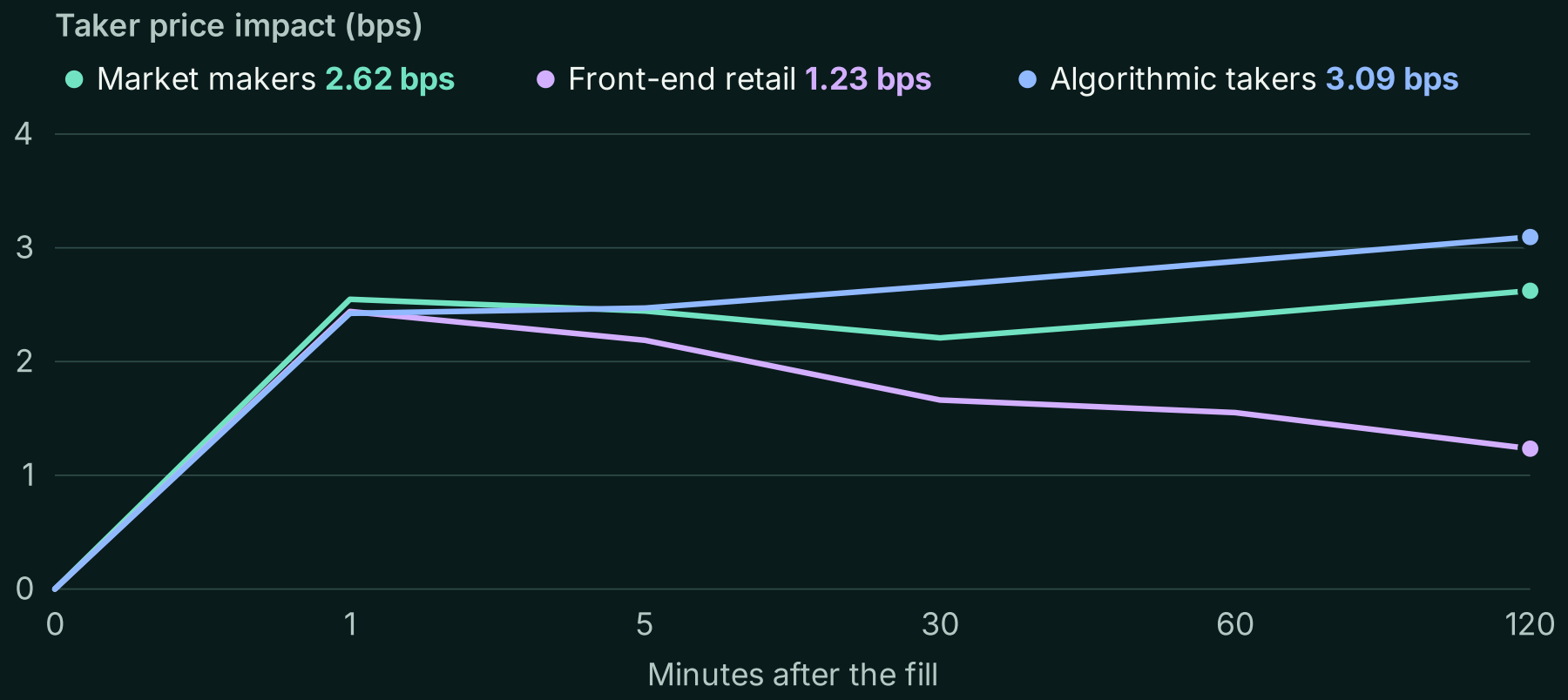
a thick book

loc limit order: size and a price cap

ASKS	SIZE
100.03	$12,000
100.02	$15,000
100.01	$15,000

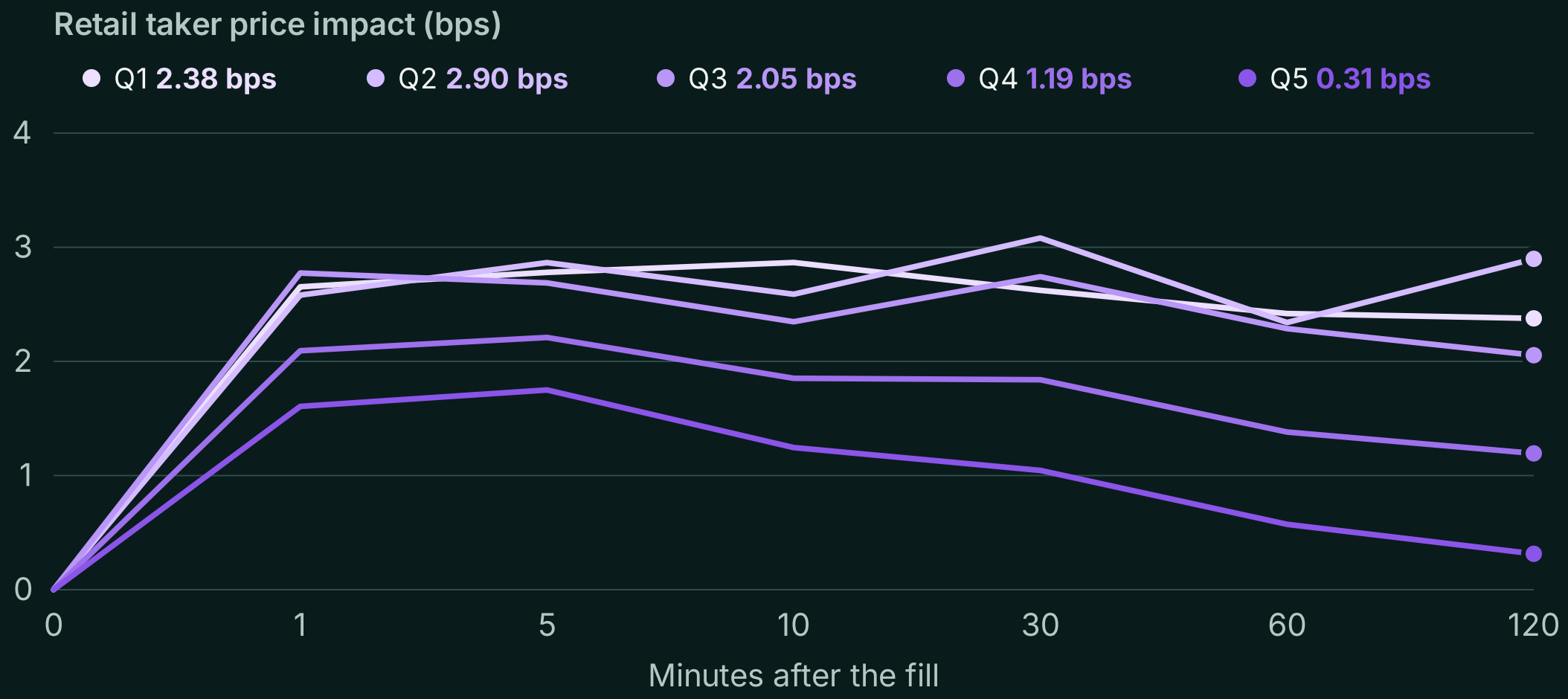
Fills in 2 levels: less slippage

Timing: Retail's Price Impact Halves Within Two Hours



- **Retail chase recent moves**, and the price reverts after them.

Timing: Price Impact Declines Most Sharply at High Leverage



Every Cohort Loses, Each for a Different Reason

- **Execution cost is the same everywhere;** price impact falls with leverage.
- **Low leverage loses through liquidation,** mostly on October 10.

COHORT	EXECUTION COST (BPS)	PRICE IMPACT, 2 H (BPS)	LIQUIDATION	FEE (BPS)	TOTAL
Q1	−2.38	**+2.38**	**−$66.8M**	−2.87	−$90M
Q2	−2.51	+2.90	−$77.0M	−2.83	−$107M
Q3	−2.70	+2.05	−$44.3M	−2.93	−$112M
Q4	−2.17	+1.19	−$33.1M	−3.19	−$107M
Q5	−2.06	**+0.31**	−$6.6M	**−3.71**	−$117M

Two Levers Would Cut Most of the Loss

Retail lose, but protecting them is not that hard

- **Fees:** a cap on trading fees would cut the largest channel.
- **Leverage limits:** the 2010 U.S. retail FX cap cut high-leverage losses by 40%.

Thank you

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